Probability

October 28, 2026

The bill this course has been running

Since Unit 1 we have written things like this:

- estimate $\pm 2\,\widehat{\mathrm{SE}}$, and called the result a 95% interval
- $\widehat{\mathrm{SE}} = s/\sqrt{n}$, with the $\sqrt{n}$ unexplained
- a p-value, described as how often you would see this by chance
- "assume the observations are independent"

Each of those is a probability statement.

Principle: this unit is machinery, not modelling

No models for three days. The payoff is that everything in Units 1 through 6 stops being a convention you follow and becomes arithmetic you can redo.

What this unit gives you

You will be able to

- read a joint table and get any probability out of it
- say whether multiplying two probabilities is legitimate
- reverse a conditional, and know why the direction matters
- compute a mean and a standard deviation from a distribution you wrote down
- derive the $\sqrt{n}$ instead of quoting it

Mistakes this prevents

- multiplying probabilities of things that share a cause
- reading $\mathbb{P}(A \mid B)$ as if it were $\mathbb{P}(B \mid A)$
- planning on the average of an uncertain input
- adding standard deviations
- trusting $n > 30$

One table for the whole unit

A roadside assistance company is called out. What the call costs them:

What happened	Cost	Share of calls
Fixed over the phone	$0	45%
Simple part	$40	30%
Major part	$150	18%
Tow and replace	$600	6%
Total loss	$2,500	1%

Definitions

- An **experiment** is any process with an uncertain outcome: a service call, a customer visit, a delivery route.
- The **sample space** is everything that could happen. For the table above it is five things.
- An **event** is a subset of it: "the call cost more than \$100" is the last three rows.
- $\mathbb{P}(A)$ is a number in $[0, 1]$ saying how often A happens if the process runs many times.

Principle: most probability errors are not arithmetic errors

They are errors about which set of outcomes is being discussed. Write the set down before you compute anything.

Rules

1. **Complement.** $\mathbb{P}(\text{not } A) = 1 - \mathbb{P}(A)$.
2. **Addition.** $\mathbb{P}(A \text{ or } B) = \mathbb{P}(A) + \mathbb{P}(B) - \mathbb{P}(A \text{ and } B)$. Subtract the overlap so it is not counted twice.
3. **Multiplication.** $\mathbb{P}(A \text{ and } B) = \mathbb{P}(A)\ \mathbb{P}(B \mid A)$. This one is *always* true. The $\mathbb{P}(B \mid A)$ is a conditional probability, defined in a moment, and it collapses to $\mathbb{P}(A)\,\mathbb{P}(B)$ only when the two are independent.

The complement rule is the one to reach for

Anything phrased "at least one" is easier backwards:

$$\mathbb{P}(\text{at least one}) = 1 - \mathbb{P}(\text{none}).$$

- A service fails on a given day with probability 0.01. Over a 90-day quarter, if the days were independent, $\mathbb{P}(\text{at least one outage}) = 1 - 0.99^{90} \approx \mathbf{0.60}$.
- A 1% daily risk is a coin flip over a quarter. Rare events stop being rare once something gets many chances at them.

Principle: count the chances, not the impressions

"That basically never happens" is a statement about one trial. Ask how many trials there are.

Conditional probability

$$\mathbb{P}(A \mid B) = \frac{\mathbb{P}(A \text{ and } B)}{\mathbb{P}(B)}$$

Throw away every outcome where B did not happen, then ask how often A happens among what is left. $\mathbb{P}(B)$ must be positive, and it is the new denominator.

- **One service call.** $\mathbb{P}(\text{cost} > \$100) = 0.18 + 0.06 + 0.01 = 0.25$. Told the call ran over \$100, the chance it was a total loss is $0.01/0.25 = \mathbf{0.04}$. That outcome started at 0.01, so the condition multiplied it by four.
- **Two tickets, drawn without replacement.** Twelve open tickets, three escalated. $\mathbb{P}(\text{first escalated}) = 3/12$. Given that, the second is escalated with probability $2/11$, so $\mathbb{P}(\text{both}) = \frac{3}{12} \times \frac{2}{11} = \mathbf{0.045}$. Not $(3/12)^2 = 0.063$, because the first draw changed the pool.
- **Same rule, rearranged.** $\mathbb{P}(A \text{ and } B) = \mathbb{P}(B)\ \mathbb{P}(A \mid B)$ is the multiplication rule written the other way round. Conditional probability is not a second idea, it is the third rule solved for the piece you did not have.

A joint table and its margins

500 delivery routes, by whether the route was downtown and whether it rained.

	no rain	**rain**	**total**
Not downtown	0.540	0.132	0.672
Downtown	0.260	0.068	0.328
total	0.800	0.200	1.000

- The four inside numbers are the **joint** probabilities. They sum to 1.
- The edges are the **marginals**, and they are what you get by summing across a row or down a column, which is where the name comes from.
- Everything else in this section is a question you can answer off this table.

Conditioning on the table

Every conditional off a joint table is one inside number over one margin.

- $\mathbb{P}(\text{downtown} \mid \text{rain}) = 0.068/0.200 = \mathbf{0.34}$.
- $\mathbb{P}(\text{downtown} \mid \text{no rain}) = 0.260/0.800 = \mathbf{0.325}$.
- $\mathbb{P}(\text{rain} \mid \text{downtown}) = 0.068/0.328 = \mathbf{0.21}$. Same numerator, a different denominator, a different question.
- The first two are nearly the same number, so knowing the weather barely moves the chance a route was downtown. That is the next slide's subject.

Principle: a probability without a condition is incomplete

Every model in this course computes a conditional: given these predictors, what do we expect. "What is the churn rate" is a worse question than "what is the churn rate for accounts under $500 a month in their first year."

Independence stated carefully

A and B are **independent** when knowing one tells you nothing about the other:

$$\mathbb{P}(A \mid B) = \mathbb{P}(A)\,, \quad \text{equivalently} \quad \mathbb{P}(A \text{ and } B) = \mathbb{P}(A)\,\mathbb{P}(B)\,.$$

- Check it on the table. $\mathbb{P}(\text{downtown})\,\mathbb{P}(\text{rain}) = 0.328 \times 0.200 = 0.0656$, against a joint of 0.068. The gap is well inside what 500 routes can resolve, so independence is plausible here.
- It should be. The weather does not consult the dispatcher. Two mortgages in one city defaulting is a different story, and so is two servers in one rack failing.

Mutually exclusive is not independence

These get used interchangeably and they are nearly opposites.

	Mutually exclusive	**Independent**
Definition	$\mathbb{P}(A \text{ and } B) = 0$	$\mathbb{P}(A \text{ and } B) = \mathbb{P}(A)\,\mathbb{P}(B)$
In words	they cannot both happen	one says nothing about the other
An example	downtown, not downtown	downtown, rain

- Mutually exclusive events are about as *dependent* as two events can be. Learning the route was downtown tells you for certain it was not suburban.
- Two events with real probabilities cannot be both: a zero joint cannot equal a positive product.

Quick check

Quick check: is the multiplication allowed?

Using the route table, a dispatcher wants $\mathbb{P}(\text{downtown and rain})$ and does not have the table in front of her. She multiplies 0.328×0.200 and gets 0.0656.

Two questions. Was she entitled to multiply? And in which of these would the same move be a disaster: (a) two routes on the same morning both running late, (b) one route being downtown and also being longer than 50 miles?

Quick check

Quick check: is the multiplication allowed?

Was she entitled to multiply, and where would the same move be a disaster?

Quick check

Quick check: is the multiplication allowed?

Was she entitled to multiply, and where would the same move be a disaster?

- **Here, yes.** The table gives 0.068 and she got 0.0656. She was lucky in the sense that she did not check, but the independence is real, because weather and route assignment have no shared cause.
- **(a) is a disaster.** Two routes on the same morning share the weather, the traffic, the depot, and the dispatcher. Multiplying treats a bad morning as two unrelated coincidences.
- **(b) is also wrong**, for a plainer reason. Downtown routes are short. The two are dependent, and here you could see it coming from what the variables mean.

Principle: independence is a claim about mechanism

You do not establish it by checking that two numbers are close. You establish it by arguing that nothing connects the two, and then the numbers agree.

Why false independence is expensive

- In 2008, models priced pools of mortgages by treating individual defaults as nearly independent. Under that assumption the chance that a large fraction defaulted at once was effectively zero.
- Defaults were driven by a common factor, the national housing market. When it turned, they turned together.
- The models were not wrong about any single mortgage. They were wrong about the **dependence**, and that was enough.

Trap: independence is an assumption, not a fact

Multiplying small probabilities makes catastrophes look impossible. That is exactly when to ask whether the events share a driver.

Your turn

Your turn: would you bet the company on it?

An engineer says: "Each of our three data centers has a 1% chance of going down in a given month, so the chance all three are down at once is 0.01^3, one in a million. We are fine."

Turn to a neighbour. What would have to be true for that number to be right, and would you sign off on it?

Your turn

Your turn: would you bet the company on it?

"One in a million, so we are fine."

Your turn

Your turn: would you bet the company on it?

"One in a million, so we are fine."

- The calculation needs the three outages to be independent. They are not. A bad deploy, an expired certificate, a provider outage, a routing misconfiguration, or an attack hits all three at once.
- One in a million is a **lower bound under the most optimistic assumption available**, and the real risk is dominated by the shared causes, none of which are in the calculation.

Principle: correlated failures are the ones that hurt

Building redundancy is a bet on independence. Spend the effort making the failures independent, not on a fourth copy of the same thing behind the same load balancer.

Conditioning Rule

$\mathbb{P}(A \mid B)$ and $\mathbb{P}(B \mid A)$ are different, often wildly different.

- $\mathbb{P}(\text{speaks Spanish} \mid \text{lives in Spain})$ is near 1. $\mathbb{P}(\text{lives in Spain} \mid \text{speaks Spanish})$ is small, because most Spanish speakers live elsewhere.
- $\mathbb{P}(\text{rich} \mid \text{owns a jet})$ is near 1. The reverse is not.
- We already have the pair off the route table: 0.21 one way, 0.34 the other, from the same 0.068 over two different margins.

Trap: the prosecutor's fallacy

"The chance of this DNA match if he were innocent is one in a million, so there is a one in a million chance he is innocent." Those are different conditionals, and the second also depends on how many people could have been the source.

A conviction built on a flipped conditional

Setup. Sally Clark was convicted in the UK in 1999 after two of her infants died suddenly. An expert testified that the chance of two cot deaths in one family was one in 73 million, computed by squaring a single-death rate.

- **First error.** The two deaths were treated as independent, which ignores shared genetics and a shared household.
- **Second error.** Even taken at face value, that number is $\mathbb{P}(\text{evidence} \mid \text{innocent})$, not $\mathbb{P}(\text{innocent} \mid \text{evidence})$. The alternative, a mother murdering two infants, is also extremely rare, and the comparison between the two is what matters.

The conviction was overturned in 2003, and the Royal Statistical Society issued a public statement on the misuse of the figure.

Bayes' rule

$$\mathbb{P}(A \mid B) \;=\; \frac{\mathbb{P}(B \mid A)\,\mathbb{P}(A)}{\mathbb{P}(B)}$$

- It is the formula for reversing a conditional, which is the operation you need constantly and the one intuition is worst at.
- $\mathbb{P}(A)$ is the **prior**, what you believed beforehand. $\mathbb{P}(A \mid B)$ is the **posterior**, what you believe after seeing B.
- Unit 10 turns this into a way of working. Here it is a tool for getting the direction right.

Principle: evidence moves a starting point, it does not replace one

A positive test shifts your belief. Where it shifts *to* depends on where it started, and the starting point is the number people leave out.

The four numbers a test comes with

Any yes-or-no test has two error rates, and the names come up constantly.

Name	The conditional	In words
Sensitivity	$\mathbb{P}(\text{test}^+ \mid \text{sick})$	of those who have it, the share the test catches
Specificity	$\mathbb{P}(\text{test}^- \mid \text{healthy})$	of those who do not, the share it clears
Prevalence	$\mathbb{P}(\text{sick})$	how common it is to begin with
Predictive value	$\mathbb{P}(\text{sick} \mid \text{test}^+)$	**what the patient wants to know**

Principle: the first three are given, the fourth is the answer

Sensitivity and specificity describe the test. The predictive value moves with the prevalence, which is why one test means different things in two populations.

Forget the formula. Count 10,000 people.

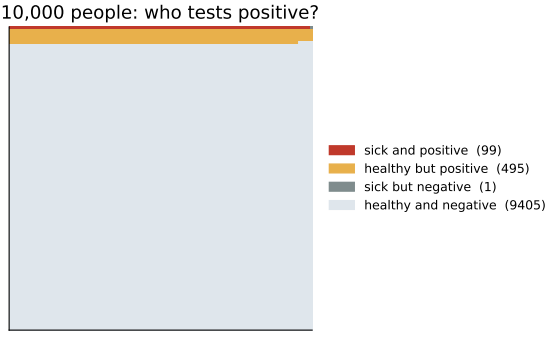
A condition affects **1%** of people. The test catches **99%** of those who have it and is wrong on **5%** of those who do not. You test positive. What is the chance you have it?

Picture 10,000 people:

- 100 have it. Of those, **99** test positive.
- 9,900 do not. Of those, **495** test positive anyway.
- So 594 people test positive, and 99 of them have it.

$$\mathbb{P}(\text{sick} \mid \text{positive}) = \frac{99}{99 + 495} \approx \mathbf{17\%}.$$

The same arithmetic as a picture



The healthy group is so much larger that its 5% error rate produces **five times** as many positives as the condition itself.

A test that is 99% accurate on a rare condition is mostly a false-positive machine. This is also your churn model, your fraud flag, and your spam filter: flag 500 accounts for a 2% outcome and most of that list is not churning. Not a broken model, just what a rare outcome does to any imperfect classifier.

Your turn

Your turn: run the numbers

A drug test is 98% accurate in both directions. It catches 98% of users and clears 98% of non-users. In a workplace where **0.5%** of employees use the drug, an employee tests positive.

Take 10,000 employees. How many test positive, and what share of them are users?

Your turn

Your turn: run the numbers

98% accurate both ways, 0.5% of employees use.

Your turn

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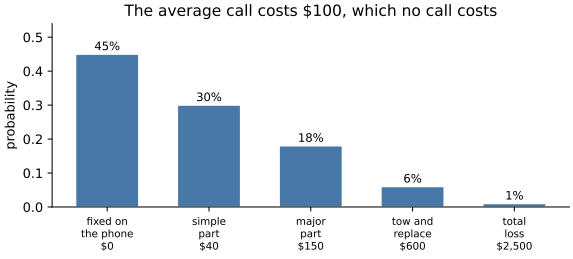
Out of 10,000 employees:

- 50 are users, and 98% of them, about **49**, test positive.
- 9,950 are not, and 2% of them, **199**, test positive anyway.
- Positives: $49 + 199 = 248$, of whom 49 are users. About **20%**.

Trap: a defensible test and an indefensible decision

Firing everyone who tests positive fires four innocent people for every user. The test is fine. Using it by itself as a decision rule is not, and the fix is usually a second test on the positives, which starts from a 20% base rate instead of 0.5%.

A random variable is the whole recipe



A **random variable** is not the number that came up. It is the full list of what is possible together with how often each happens.

Write X for the cost of one service call. The bars are $\mathbb{P}(X = x)$. Nothing here is fitted or estimated. We wrote the distribution down, and now we can ask it questions.

Expectation off the table we built

$$\mathbb{E}[X] \;=\; \sum_{x} x \cdot \mathbb{P}(X = x)$$

$$(0)(0.45) + (40)(0.30) + (150)(0.18) + (600)(0.06) + (2500)(0.01)$$

$$= 0 + 12 + 27 + 36 + 25 = \$\mathbf{100}$$

- It is the average over a very large number of calls, and it need not be a possible value. No call costs \$100.
- The **median** call costs \$40, and 45% of calls cost nothing at all. The average is not the typical call, and both numbers are correct answers to different questions.

Principle: expectation answers "in the long run"

It is the right tool for a decision you make thousands of times and a dangerous one for a decision you make once with everything on the line.

The tail owns the mean

Look at where the $100 came from:

Outcome	Cost	Share	Contribution to the mean
Fixed over the phone	$0	45%	$0
Simple part	$40	30%	$12
Major part	$150	18%	$27
Tow and replace	$600	6%	$36
Total loss	$2,500	1%	**$25**

- A quarter of the average cost comes from an outcome that happens once in a hundred calls.
- Three quarters of calls cost $40 or less and together account for $12 of the $100.

In practice: what this means for your data

A mean carried by a rare row partly estimates how often that row happens.

Linearity is the fact to lean on

For *any* random variables, independent or not:

$$\mathbb{E}[aX + b] = a\mathbb{E}[X] + b, \qquad \mathbb{E}[X + Y] = \mathbb{E}[X] + \mathbb{E}[Y]\,.$$

- No independence needed anywhere. This is why expectations are pleasant and variances are not.
- **Use it.** The company takes 400 calls a day. Daily cost is $T = X_1 + \cdots + X_{400}$, so $\mathbb{E}[T] = 400 \times \$100 = \$\mathbf{40{,}000}$, whether or not busy days cluster.
- We never needed the distribution of the daily total to get its mean. Linearity handed it over.

Variance and why we report its square root

$$\mathrm{Var}(X) = \mathbb{E}\big[(X - \mathbb{E}[X])^2\big], \qquad \mathrm{SD}(X) = \sqrt{\mathrm{Var}(X)}.$$

- Variance is in squared units. Dollars squared means nothing to anyone, so report the **standard deviation**, which is back in dollars.
- The version that is easier to compute by hand: $\mathrm{Var}(X) = \mathbb{E}\big[X^2\big] - (\mathbb{E}[X])^2$.
- $\mathrm{Var}(aX + b) = a^2\mathrm{Var}(X)$. Shifting does nothing to spread. Scaling squares.

Principle: the mean and the spread answer different questions

"What will this cost on average" and "how bad can a month get" are both reasonable, and only the second one needs the variance.

Variance by hand on the service call

The \$0 outcome contributes nothing, so four terms are left:

$$\begin{aligned}\mathbb{E}\big[X^2\big] &= (1{,}600)(0.30) + (22{,}500)(0.18) + (360{,}000)(0.06) + (6{,}250{,}000)(0.01)\\ &= 480 + 4{,}050 + 21{,}600 + 62{,}500 \;=\; 88{,}630\end{aligned}$$

$$\mathrm{Var}(X) = 88{,}630 - 100^2 = 78{,}630, \quad \mathrm{SD}(X) = \sqrt{78{,}630} = \$\mathbf{280}$$

- The standard deviation is nearly three times the mean. The 1% outcome, squared, supplies \$62,500 of the \$88,630.
- A distribution where the spread dwarfs the centre is not unusual in business data. It is what costs, claims, revenues, and durations usually look like.

Variances add. Standard deviations do not.

For **independent** X and Y:

$$\mathrm{Var}(X+Y) = \mathrm{Var}(X) + \mathrm{Var}(Y), \quad \text{so} \quad \mathrm{SD}(X+Y) = \sqrt{\mathrm{Var}(X) + \mathrm{Var}(Y)}.$$

- Across 400 independent calls, $\mathrm{Var}(T) = 400 \times 78{,}630 = 31{,}452{,}000$ and $\mathrm{SD}(T) = \sqrt{400} \times \$280 = \$\mathbf{5{,}608}$.
- Not $400 \times \$280 = \$112{,}000$. Spread grows like $\sqrt{n}$, and that single fact is why averages are worth anything at all.

Trap: the independence clause, again

"Variances add" is not a law of nature. It is a reward for independence, and it is the same assumption that failed in 2008. If busy days make every call harder, the variance of the total is larger than this.

Quick check

Quick check: budget the bad day

The daily total has mean \$40,000 and standard deviation \$5,608.

A manager reasons: "each call swings by \$280, and I have 400 calls, so the day could easily be \$112,000 off." Where did that go wrong, and roughly what is the right order of magnitude for how far off a day can be?

Quick check

Quick check: budget the bad day

"Each call swings $280, 400 calls, so $112,000 off."

Quick check

Quick check: budget the bad day

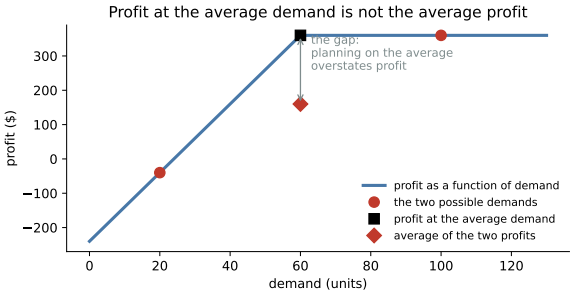
"Each call swings \$280, 400 calls, so \$112,000 off."

- He added standard deviations. That is only right if every call goes wrong together, which is the worst case and not the usual one.
- Variances add, so the standard deviation of the total is $\sqrt{400} \times \$280 = \$5{,}608$, twenty times smaller than his number.
- So a day that is \$10,000 over is unremarkable and a day that is \$100,000 over is not a bad day, it is a different business. Unit 8 attaches a probability to each.

Principle: the reason a big operation is predictable

Individual calls are wildly unpredictable, with a spread three times their mean. The daily total is predictable to within about 14%. Nothing changed except how many of them there are.

Curved functions break the average



Capacity is 60 units, each sold for \$10, each unit of capacity costs \$4. Demand is 20 or 100, equally likely.

- Profit at the *average* demand of 60: **\$360**.
- *Average* profit across the two demands: **\$160**.

The plan built on the average overstates profit by more than double, and no arithmetic error was made anywhere.

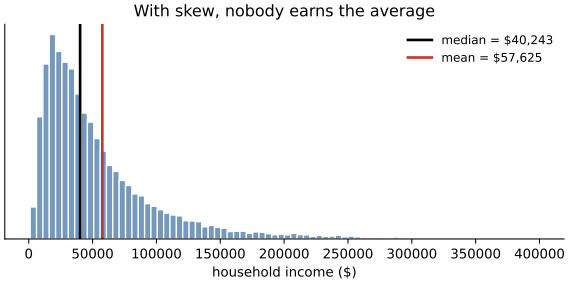
Why the gap appears

- Whenever the thing you care about is a **curved** function of an uncertain input, $\mathbb{E}[f(X)] \neq f(\mathbb{E}[X])$.
- Caps, floors, thresholds, penalties, stockouts, and service-level cliffs are all curves. Operations are made of them.
- **It can cut your way.** Suppose the company charges a flat \$120 per call and caps its own exposure at \$500. Expected payout is $\mathbb{E}[\min(X, 500)] = \$74$, not \$100, so profit is **\$46** a call rather than the \$20 you get from $120 - \mathbb{E}[X]$. The cap is worth \$26 a call, and it is invisible if you plan on the mean.

Trap: planning on the average

Replace every uncertain input with its mean and the plan will be systematically wrong. Which direction depends on which way the curve bends, so you have to look.

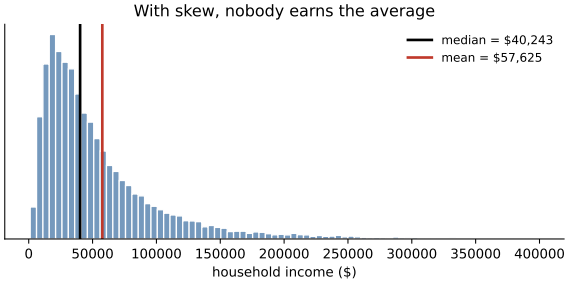
The mean is not the typical case



With right-skewed data the mean sits well above the median, pulled up by the tail. Our service call does the same thing: median $40, mean $100.

- "Average revenue per user" in a business where 2% of users are whales describes nobody.
- Report the median, the mean, and something about the tail. They answer different questions and the gap between them is itself information.

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- "Average revenue per user" in a business where 2% of users are whales describes nobody.
- Report the median, the mean, and something about the tail. They answer different questions and the gap between them is itself information.

When the mean and the median are far apart, somebody is about to be misled by one of them.

Conditioning an expectation

Conditional probability shrank the sample space. Conditional expectation takes the average inside what is left.

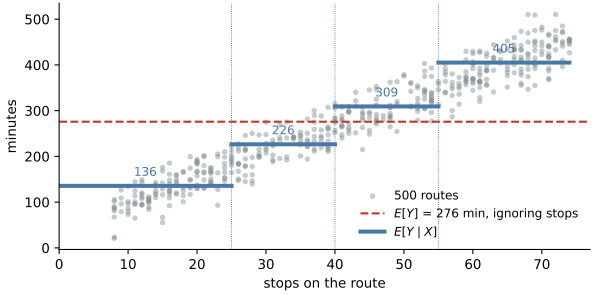
$$\mathbb{E}[Y \mid X = x] \;=\; \sum_{y} y \cdot \mathbb{P}(Y = y \mid X = x)$$

- Same formula as before, computed on the rows where $X = x$ instead of on all of them.
- $\mathbb{E}[Y]$ is one number. $\mathbb{E}[Y \mid X]$ is a **function of** X: a different answer for each value you condition on.

Principle: this is what every model in this course reports

A fitted value is a conditional expectation. When Unit 3 predicted rent from size, it was estimating $\mathbb{E}[\text{rent} \mid \text{size}]$, and every argument in that unit was about doing it well.

Four numbers instead of one



500 delivery routes. Ignoring stops, the best single guess for a route is the overall mean, **276 minutes**, the dashed line.

Condition on how many stops the route has and you get four numbers instead: **136, 226, 309, 405**. Each is an average taken inside its own group.

The blue steps are $\mathbb{E}[Y \mid X]$. Nothing was fitted. These are just four averages.

The overall mean is a weighted average of the pieces

$$\mathbb{E}[Y] \;=\; \sum_{x} \mathbb{P}(X = x)\;\mathbb{E}[Y \mid X = x]$$

Stops	**Share of routes**	$\mathbb{E}[\text{minutes} \mid \text{stops}]$	**Contribution**
1 to 25	0.276	135.6	37.4
26 to 40	0.200	226.5	45.3
41 to 55	0.200	309.3	61.9
56 and up	0.324	405.1	131.2
			275.8

That total is the overall mean, 275.8 minutes, exactly. Conditioning rearranges an average into groups. It never changes the average.

This is what a regression tree computes

- A tree splits the rows into groups and reports the mean of each group. That is $\mathbb{E}[Y \mid X]$ for the groups the splitting happened to find.
- The step function in the figure *is* a depth-2 tree on `stops`, with the cut points chosen by hand instead of by the greedy search of Unit 5.
- So the thing a tree is estimating has a name, and you now have it. A regression line estimates the same conditional expectation with a straight line instead of steps.

Principle: the models differ in the shape, not in the target

Linear regression, a tree, and a forest are all estimating $\mathbb{E}[Y \mid X]$. They disagree about what shape that function is allowed to take, which is what Units 3 through 6 were arguing about.

Quick check

Quick check: recover the overall mean

A colleague has the four conditional means (136, 226, 309, 405) and reports the overall average route time as their simple average, $(136 + 226 + 309 + 405)/4 = 269$ minutes.

The right answer is 276. What did he do wrong, and what would have to be true for his method to work?

Quick check

Quick check: recover the overall mean

He averaged the four group means and got 269 instead of 276.

Quick check

Quick check: recover the overall mean

He averaged the four group means and got 269 instead of 276.

- He weighted the four groups equally. They are not equal: the biggest group is 32% of routes and the smallest is 20%.
- The weights have to be the group *shares*, which is what the law of total expectation says. His method is right only when every group is the same size.
- His error is small here, 7 minutes, because the groups are not far from equal. Build the same table with a group that is 2% of the data and the error is enormous.

Trap: averaging averages

"Average the regional averages" is the most common version of this, and it quietly gives a hundred-customer region the same say as a hundred-thousand-customer region.

Why averaging works

Write $\bar{X}$ for the average of n independent draws. Everything needed is already proved:

$$\mathbb{E}\big[\bar{X}\big] = \frac{1}{n}\left(\mathbb{E}[X_1] + \cdots + \mathbb{E}[X_n]\right) = \mu \qquad \text{linearity, no independence needed}$$

$$\mathrm{Var}\big(\bar{X}\big) = \frac{1}{n^2}\left(\mathrm{Var}(X_1) + \cdots + \mathrm{Var}(X_n)\right) = \frac{\sigma^2}{n} \qquad \text{variances add, independence needed}$$

$$\mathrm{SD}\big(\bar{X}\big) = \frac{\sigma}{\sqrt{n}}$$

Principle: there is the $\sqrt{n}$

It came from two lines: scaling squares inside a variance, and variances add for independent things. That is all of it, and it has been sitting under every interval in this course since Unit 1.

Standard error is not standard deviation

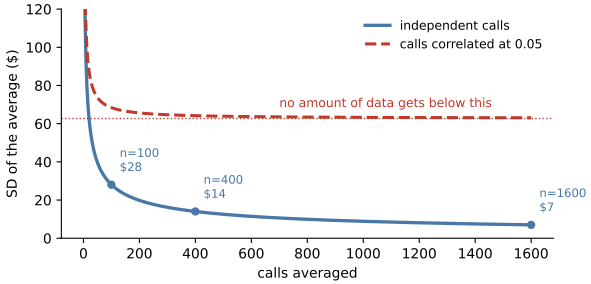
	Standard deviation	**Standard error**
Measures	spread of one call	spread of an average of calls
For our table	\$280	$\$280/\sqrt{n}$
At $n = 400$	\$280	\$14
Shrinks with more data	no	yes

- The standard deviation is a fact about the calls. More data measures it better but does not change it.
- The standard error is a fact about your *estimate*. It shrinks because you have more of them.

Trap: the confusion that produces absurd intervals

Reporting $\pm 2 \times \$280$ when you meant $\pm 2 \times \$14$, or the reverse. Ask which one the question is about: one call, or the average of many.

What four times the data buys



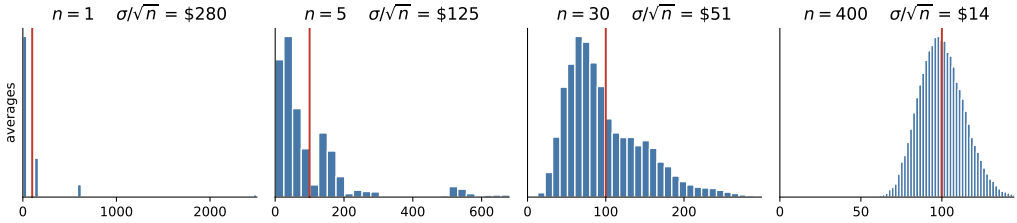
From 100 calls to 400 cuts the standard error from \$28 to \$14. From 400 to 1,600 cuts it to \$7.

Four times the data, half the error. That is the exchange rate, and it is why the second half of a data collection effort is worth so much less than the first.

The dashed line is the same calculation when calls are correlated at only 0.05. It flattens onto a floor, and no sample size gets below it.

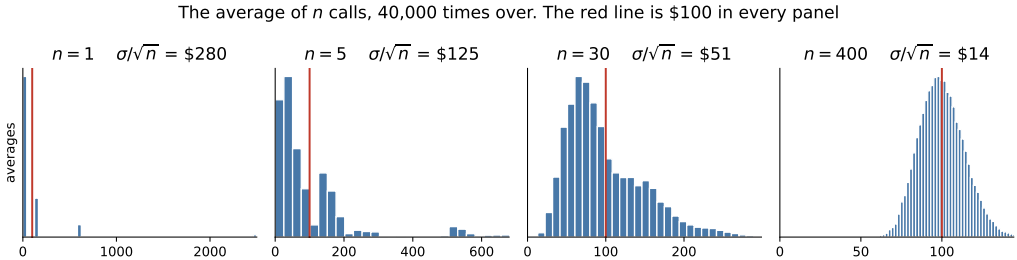
The shape settles down too

The average of n calls, 40,000 times over. The red line is \$100 in every panel



At $n = 1$ it is the five spikes we started with. By $n = 400$ it is one smooth hump centred on \$100, and the spread has shrunk exactly as derived: \$280, \$125, \$51, \$14.

The shape settles down too



At $n = 1$ it is the five spikes we started with. By $n = 400$ it is one smooth hump centred on \$100, and the spread has shrunk exactly as derived: \$280, \$125, \$51, \$14.

This is the **central limit theorem**. Averages of independent draws settle into one particular bell shape no matter what was averaged. Unit 8 names that shape and computes with it.

Thirty is not a magic number

- Look again at the $n = 30$ panel. It is still visibly leaning right, because 1% of calls cost \$2,500 and a sample of 30 either contains one of those or it does not.
- Measured skewness of the average: **6.6** at $n = 1$, **1.2** at $n = 30$, **0.3** at $n = 400$. Thirty got most of the way and not all of it.
- How large n has to be depends entirely on how skewed the thing being averaged is. For a coin flip, 30 is plenty. For anything with a rare expensive tail, it is not.

Trap: $n > 30$

It is a rule of thumb for mildly skewed data that got promoted to a law. The honest version: the more the tail matters, the larger n has to be, and you can always check by simulating.

Four things this does not say

1. **It does not say the data becomes normal.** The service calls are still five spikes no matter how many you collect. The *average* is what settles down.
2. **It does not help with one observation.** It says nothing about what the next call will cost.
3. **It needs independence.** Correlated draws have the floor from two slides ago, and the $\sqrt{n}$ stops applying.
4. **It needs a finite variance.** Some real distributions do not have one, and then averaging does not stabilise anything.

In practice: what to do instead of trusting it

Simulate. You have the distribution, or a sample from it, and resampling it at your actual n answers the question directly. That is Unit 6, used as a safety check on Unit 7.

Your turn

Your turn: the quarterly number

The daily cost total has mean \$40,000 and standard deviation \$5,608.

A VP wants the uncertainty on the 90-day quarterly total. She multiplies the daily standard deviation by 90 and gets \$504,720.

With a neighbour: what is the right calculation if days are independent, and why is the truth probably somewhere between the two answers?

Your turn

Your turn: the quarterly number

She multiplied the daily SD by 90 and got $504,720.

Your turn

Your turn: the quarterly number

She multiplied the daily SD by 90 and got $504,720.

- **If days are independent**, variances add across days, so $\mathrm{SD}(\text{quarter}) = \sqrt{90} \times \$5{,}608 = \$\mathbf{53{,}200}$. Her answer is about ten times too large.
- **Days are not independent.** A hard winter, a recall, a holiday stretch, or a price change moves every day in the quarter the same direction. That pushes the real number above $53,200.
- **Her number is the ceiling**, the answer if all 90 days moved in perfect lockstep. The truth is between $53,200 and $504,720, and which end depends on how much the days share.

Principle: quote the assumption with the number

"$53,200, assuming days are independent, and more than that to the extent they are not" is a usable answer. Either single number on its own is not.

Turning a probability question into arithmetic

1. **Write down the sample space.** What are the outcomes, and what are their probabilities. Most errors die here.
2. **Decide whether you need a joint, a marginal, or a conditional.** If the question has a "given" or an "of those," it is conditional.
3. **Check the direction.** Is the number you have $\mathbb{P}(A \mid B)$ when the question asked for $\mathbb{P}(B \mid A)$. If so, you need the base rate too.
4. **Before multiplying, say out loud why the two things are unrelated.** If you cannot, do not multiply.
5. **For means, use linearity.** It never needs independence.
6. **For spreads, add variances and take the square root at the end.** This does need independence.
7. **If a cap, floor, or threshold is involved, do not plug in the mean.** Average the outcomes instead.

What you should be able to do with software

Task	How
Its mean and variance	`(x*p).sum()` and `(x**2*p).sum() - mean**2`
Draw from it	`rng.choice(x, size=n, p=p)`
A joint table from data	`pd.crosstab(a, b, normalize=True)`
Check independence	compare the cell to the product of its margins
A conditional expectation	`df.groupby(x)[y].mean()`
Law of total expectation	weight those by `value_counts(normalize=True)`
Anything you cannot derive	simulate it at your actual n

In practice: the habit worth keeping

Unsure whether a probability argument holds? Simulate it.

The traps

- Multiplying probabilities of events that share a cause.
- Confusing mutually exclusive with independent, and multiplying when you should add.
- Reading $\mathbb{P}(A \mid B)$ as $\mathbb{P}(B \mid A)$, and leaving out the base rate.
- Quoting an accuracy for a rare outcome as though it were a predictive value.
- Adding standard deviations instead of variances.
- Planning on the mean of an input that feeds a cap, a floor, or a penalty.
- Averaging group averages without weighting by group size.
- Treating $n > 30$ as a guarantee, especially for anything with a rare expensive tail.

Ten questions to be able to answer

1. What is the sample space, and do these probabilities sum to one.
2. Give a marginal, a joint, and a conditional probability off a two-way table.
3. When does $\mathbb{P}(A \text{ and } B) = \mathbb{P}(A)\,\mathbb{P}(B)$, and how would you defend it here.
4. Why can two events with positive probability not be both mutually exclusive and independent.
5. A test is 99% accurate for a 1-in-100 condition. You test positive. Now what.
6. Compute a mean and a standard deviation from a five-row distribution.
7. Which of linearity and variance-adding needs independence, and why.
8. Why is $\mathbb{E}[f(X)] \neq f(\mathbb{E}[X])$ when f has a cap in it.
9. Derive $\mathrm{SD}(\bar{X}) = \sigma/\sqrt{n}$ from the two rules.
10. What does the central limit theorem claim, and what does it not claim.

Where we go next

- This unit wrote distributions down by hand. Unit 8 asks where one comes from when nobody hands you the table, which means picking a rule by thinking about the mechanism and then checking whether the data agrees.
- It also names the bell shape that showed up at the end of this unit, so we can finally turn a spread into a probability and answer questions like how often the quarter comes in over budget.
- And it goes back to the prediction intervals of Unit 4 to point out something we skipped: a forecast was never a single number. It was a distribution the whole time.